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Evaluation of "The Interprofessional Education Academy"

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Abstract

This study evaluated the impact of the "1st Interprofessional Education Academy" on the interprofessional identity and attitudes of healthcare students in Turkey. This was a 4 day national training program for 24 health professionals aimed to increase effective collaboration and communication skills, support the development of professional identity, and strengthen the implementation of IPE in Turkey. Twenty-four female participants (mean age: 22.88 ± 2.35 years) from various healthcare disciplines completed the Extended Professional Identity Scale (EPIS-TR) and the Interprofessional Attitude Scale (IPAS) before and after the intervention. The results showed significant improvements in all EPIS-TR subscales (interprofessional belonging, commitment, and beliefs) and total scores, indicating a positive impact on interprofessional identity. The IPAS results revealed significant increases in the teamwork, roles, and responsibilities (TRR), patient-centeredness (PC), cultural competency (CC), and overall scores, suggesting enhanced interprofessional attitudes and competencies. A moderately positive correlation was found between EPIS-TR and IPAS total scores. In the analysis, it was seen that there was an increase in the perception level of the participants regarding the importance of IPE due to the effect of the training they received. In addition, there was a significant decrease in participants' perceptions of their place in the training programs due to an increase in their awareness. The training program received positive evaluations from the participants. These findings highlight the effectiveness of structured IPE interventions in fostering collaborative attitudes and identities among health care students. Further research with larger and more diverse samples is needed to reinforce these results and provide additional evidence for the scale's applicability in IPE contexts.

Keywords Interprofessional education, Professional identity, Interprofessional identity, Interprofessional attitudes, Collaboration, Healthcare students

Introduction

Interprofessional collaboration (IPC) is a vital clinical approach aimed at improving the quality of healthcare [1–3]. This collaboration aims to provide more effective and holistic solutions for patient care by bringing together the expertise of various health professionals. The formation of multidisciplinary teams to solve complex problems in the health system improves patient outcomes, as well as the efficiency of health services and patient satisfaction [4, 5]. Through interprofessional cooperation, communication and coordination between health professionals are strengthened, paving the way for

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the provision of better quality health services [4, 6]. Internationally, many researchers are working to understand the nature of inter-professional collaboration [7–11]. Current studies suggest the development of case-based student-centered small group training activities targeting core competencies at the earliest possible stage for the development of IPC [8, 10, 12]. In these studies, many aspects of the activities, such as participants' interprofessional identity (IPI) and attitude (IPA) development, were evaluated through changes in quantitative and qualitative designs [12–14].

IPI refers to the sense of belonging, commitment, and beliefs that individuals develop as part of collaborative team-based work in healthcare, fostering mutual respect and shared goals across professional disciplines [15]. Many theories have been proposed to assess the IPI [15–17]. In the context of these theories, IPI is a complex structure shaped by the interactions and collaborations of professionals in the field of health services. This identity requires individuals to strike a balance between their professional identity and the identities of other professional groups. Interprofessional education (IPE) plays a critical role in the development of identity. Studies have shown that IPE helps health professionals develop their professional identities. For example, IPE allows students to understand both their professional identity and the roles of other professional groups [18, 19]. IPI development involves recognizing individuals' professional roles, respecting the roles of other professional groups, and learning how to interact with them. This process occurs through socialization and experience. It has been observed that health professionals develop their IPI and professional identity when working with each other [20, 21]. This type of training helps students develop a broader professional identity beyond their boundaries [22]. For IPE programs to be effective, it is necessary to create environments that enable participants to interact with one another. Research has shown that such interactions help participants better understand and develop their professional identities [23, 24]. IPE practices are important tools for promoting collaboration in healthcare [25, 26]. In this context, it is recommended to redesign training curricula and adopt skill frameworks that support IPC [2, 20]. IPI is an important component of health professionals' educational processes. The development of this identity is shaped by professional interactions and collaborations. Supporting these interactions in educational activities is critical for improving the quality of healthcare services and developing professionals' respect for each other [27, 28].

Interprofessional attitudes refer to the attitudes that health professionals develop during their interactions and collaborations with each other. This attitude is shaped by IPE (IPE) processes and plays an important role in improving the quality of health care services. Research has shown that IPE improves health professionals' ability to collaborate and build positive attitudes [29–31]. In particular, students' IPE experiences were found to strengthen their attitudes towards teamwork and collaboration [32, 33]. IPE enables healthcare professionals to interact with colleagues from different disciplines, and these interactions increase their understanding of each other's roles. Situations in which students from different departments work together have been reported to improve collaboration and communication skills [31, 34]. Such experiences positively affect students' attitudes towards interprofessional cooperation [33, 35]. However, obstacles are also encountered in interprofessional cooperation and training processes. Research has shown that factors such as a lack of time, resources, and interprofessional competition can negatively affect collaborative attitudes [34, 36]. The reluctance of some health professionals to collaborate with other professional groups may reduce the effectiveness of IPE practices [36, 37]. In this context, it is important that health professionals' training activities develop strategies that promote inter-professional collaboration [34, 38]. Interprofessional attitudes are a critical component of the education process and practices of health professionals. IPE is an important tool in developing these attitudes and enables professionals to work effectively with each other to improve the quality of health care [39, 40]. Adopting an approach that supports interprofessional cooperation in education programs will contribute to a more effective and efficient delivery of health services [41, 42].

Studies on interprofessional cooperation have emphasized the role and importance of IPE [3, 43–45]. From a psychological perspective, identity refers to a person's sense of self shaped by individual personality, experiences, and social interactions, whereas attitude is defined as organized emotions, beliefs, and behavioral tendencies [46, 47]. Social identity and attitude depend on an individual's psychological relationships with a group or social category [48]. In this context, our study was designed as a case-based small-group training in an environment where individuals can interact socially to ensure their socialization. There are many approaches to and suggestions for developing such training programs [45, 49, 50].

The "1st Interprofessional Education Academy" was held in Antalya on 10–13 September 2024 with the aim of organizing a national training program in the field of IPE in our country. The event aimed to increase effective collaboration and communication skills among health care professionals, support the development of professional identity, and strengthen the implementation of IPE in Turkey. The training program aimed to improve participants' knowledge, skills, attitudes, and awareness. The program lasted for four days and increased

the competencies of the participants in interprofessional cooperation (IPC) through different activities each day. On the first day, basic concepts related to IPE were discussed with all participants in a classroom setting to increase their level of knowledge. On the second day, case scenario-based training activities were conducted in small groups to help participants develop skills and develop positive attitudes towards IPE. On the third day, workshops and discussion sessions were organized to increase the awareness and sensitivity levels of the participants. On the last day, feedback was received, and the training process was completed with a certificate ceremony and closing speeches.

The event, attended by 10 trainers and 24 participants, consisted of three parts [51]. On the first day of the event, topics related to competencies related to inter-professional cooperation were covered. On the second day, small-group work and presentations on four scenarios and four core competencies were realized. On the third day, a workshop was held on what needed to be done at the individual, university, and national levels regarding interprofessional education/collaboration. At the end of the training, all participants were certified and a final report was prepared.

The effect of training activities on identity and attitude has been shown in the context of socialization theory [45, 49, 50]. This study aimed to evaluate the effect of a training program developed in accordance with the interprofessional collaboration competencies defined in the framework of CAIPE on interprofessional identity and attitude.

Method

This study had a quantitative research design and was designed as a pre-post evaluation using validated tools. 24 students participated face-to-face ($n = 24$). Ethics committee approval was obtained from the "Süleyman Demirel University Ethics Committee" for the study (date: 03.07.2024 No: 77/2). In this study, consent that was obtained from all of the participants was informed. In this study, the EPIS-TR and IPAS, which had previously been studied for validity and reliability in Turkish, were used for the pre-test and post-test [52, 53].

The EPIS-TR scale was developed to measure the IPI. It consists of three sub-dimensions: interprofessional belonging, commitment, and beliefs. The interprofessional belonging subscale measures an individual's sense of belonging to different professional groups and their relationships with these groups. Interprofessional belonging is an important factor that affects professionals' commitment to each other and their willingness to cooperate. The Interprofessional Commitment dimension assesses individuals' commitment to IPC and their willingness to maintain it. Interprofessional commitment is critical for improving the quality of health care services. The Interprofessional Beliefs subscale measures health professionals' beliefs about IPC and the impact of these beliefs on their professional practices. Beliefs are an important element in shaping professionals' attitudes towards collaboration. The scale consisted of 12 questions scored on a five-point Likert-type scale. The scale has no negative questions and no cutoff score [17].

The IPAS assesses the level of inter-professional attitudes. The scale consists of five subscales (teamwork, roles, and responsibilities (TRR); patient-centeredness (PC); interprofessional bias (IPB); diversity and ethics (DE); and community centeredness (CC)). The Teamwork, Roles, and Responsibilities (TRR) subscale measures health professionals' attitudes towards teamwork and the roles and responsibilities of different professional groups. Teamwork is critical for effective delivery of health services. The Patient-Centeredness (PC) dimension assesses health professionals' attitudes towards adopting a patient-centered approach. Patient-centered service delivery is an important element for improving the quality of healthcare services. IPB subscale measures the biases that health professionals may have against other professional groups and the impact of these biases on IPC. Thus, reducing interprofessional prejudice is essential for effective collaboration. DE dimension assesses health professionals' attitudes towards diversity and ethical values. Diversity and ethics are important for ensuring justice and equity in health care. CC subdimension measures health professionals' understanding of community-oriented services and their sensitivity to community needs. A community-centered approach can increase the effectiveness of health services. The scale consists of 27 questions scored on a seven-point Likert-type scale. This scale, which has no cutoff score and no negative questions, is widely used in the literature for attitude assessment [52, 54–58].

Within the scope of this study, an important performance analysis of the training program was conducted. The participants were asked to evaluate "the importance of IPE in your profession" and "the performance of IPE" in your profession (its place in your education)".

Finally, participants' satisfaction with the activity was evaluated.

JAMOVI and SPSS software were used for statistical analysis. Means and standard deviations were calculated for participants' age, scale scores, and satisfaction scores. For confirmatory tests between groups, the T-Test or Mann-Whitney U test was performed according to the distribution in the dependent groups (Shapiro-Wilk). An importance performance analysis was performed to determine the content of the training program.

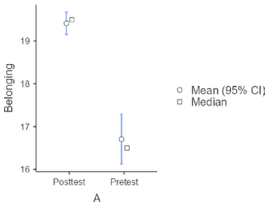
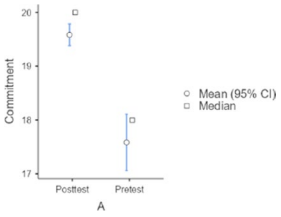
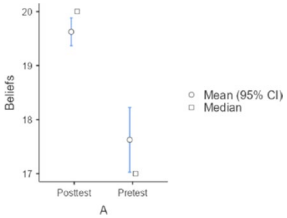
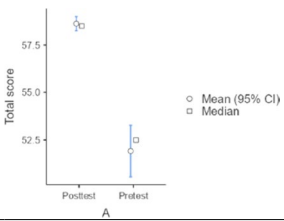
Results

All participants were female, and the mean age was 22.88 ± 2.35 years. (minimum: 20 years; maximum: 31 years). As the study was designed as a project, an equal number of students from different professions were accepted. Therefore, six professions (nursing, midwifery, medicine, disaster management, nutrition and dietetics, and dentistry) were represented by four students, comprising 16.66% of the total participants ($n = 24$).

According to the results of the Shapiro-Wilk normality test, the EPIS-TR data were not normally distributed. For interprofessional belongingness, the W value was 0.94, the p-value was 0.015, the pre-test mean was 16.71 ± 1.46, and the post-test mean was 19.42 ± 0.65 ($p < 0.001$, d: 2.40). For Interprofessional Commitment, W value was 0.81, p-value was < 0.001, pre-test mean was 17.58 ± 1.32, and post-test mean was 19.58 ± 0.50 ($p < 0.001$, d = 0.91). For Interprofessional Beliefs, the W value was 0.90, the p-value was < 0.001, the pre-test mean was 17.63 ± 1.50, and the post-test mean was 19.63 ± 0.65. A significant difference was observed at $P < 0.001$ (d = 0.75). The W-value for EPIS-TR total scores was calculated as 0.88, p-value < 0.001, the pre-test mean was 51.92 ± 3.40, and the post-test mean was 58.63 ± 0.92 ($p < 0.001$, d = 1.00) (Table 1).

According to the results of the Shapiro-Wilk normality test, it was observed that the IPAS data were not normally distributed. For TRR, the W value was 0.94, the p-value was 0.015 and the pretest mean was 53.25 ± 4.11 and the posttest mean was 58.96 ± 3.42. ($p < 0.001$, Cohens d: 0.70). The W-value for PC was 0.80, p-value < 0.001). The pre-test mean was 32.17 ± 3.61, and the post-test mean was 33.67 ± 2.18 ($p = 0.048$, d = 0.31). For IB, the W-value was 0.93, the p-value was 0.007, the pre-test mean was 14.21 ± 4.12, and the post-test mean was 15.17 ± 3.81 ($p = 0.403$, d: 0.14). For DE, the W value was 0.71, p-value < 0.001, pretest mean was 26.92 ± 1.74, and post-test mean was 27.75 ± 0.68 ($p = 0.066$, d = 0.23). For CC, the W value was calculated as 0.78, p-value < 0.001, the pre-test mean was 37.75 ± 5.24, and the post-test mean was 41.25 ± 1.42 ($p = 0.004$, d = 0.44). The W-value for the IPAS total score was 0.93, and the p-value was 0.005. The

Table 1 EPIS-TR analyses

	Normality Test (Shapiro-Wilk)		EPIS-TR				
	W	*p*	Pre-test	Post test		*p*	Effect Size
Interprofessional Belonging	0.94	0.015	16.71 ± 1.46	19.42 ± 0.65		< 0.001*	2.40
Interprofessional Commitment	0.81	< 0.001	17.58 ± 1.32	19.58 ± 0.50		< 0.001**	0.91
Interprofessional Beliefs	0.90	< 0.001	17.63 ± 1.50	19.63 ± 0.65		< 0.001**	0.75
Total score	0.88	< 0.001	51.92 ± 3.40	58.63 ± 0.92		< 0.001**	1.00

*Student's t

**Mann-Whitney U

pre-test mean score was 164.29 ± 12.47, and the post-test mean score was 176.79 ± 6.70 ($p < 0.001$, d: 0.65). These findings show that there were significant differences in the TRR, PC, CC, and total scores, while the differences were not significant for the IB and DE variables (Table 2).

The Pearson correlation coefficient between EPIS-TR and IPAS total scores was found to be 0.44 and statistically significant at $p < 0.01$ level. This finding indicates a moderately positive relationship between the two variables. It was concluded that, as EPIS-TR scores increased, IPAS scores also increased, and this relationship was significant. The correlation coefficient of 0.44 indicates that there is a significant positive relationship between EPIS-TR and IPAS. The p-value was 0.002, which confirmed that the relationship was statistically significant at the 1% significance level.

In our study, participants were asked to evaluate the concept of "IPE" in terms of "your opinion, its importance in your profession" and the concept of "IPE" in terms of "its performance in your profession (its place in your education)" at the beginning and end of the training according to the content of the training (Item 1: the importance of IPE in our country, Item 2: the importance of IPE in our university, Item 3: IPE identity theory, Item 4: IPE roles and responsibilities, Item 5: IPE teamwork in health professionals, Item 6: IPE values and ethics, Item 7: IPE communication, and Item 8: International IPE organizations). Participants were asked to score these activities out of 5 points. In the analysis, it was seen that there was an increase in the perception level of the participants regarding the importance of IPE due to the effect of the training they received. In addition, as the participants' awareness of IPC increased, there was a significant decrease in their perception of the place of IPC in their professional training programs (Fig. 1).

As a result of the evaluation of the training, the organization of the training was found to be positive, with a score of 4.33 ± 0.69. Training materials were evaluated as adequate with 4.13 ± 1.01. Courses have a high satisfaction rate of 4.38 ± 0.81 points. Small-group training was positively received by the participants with a score of 4.04 ± 0.93, but slightly more variable satisfaction was reported compared to the other components. With a score of 4.17 ± 0.80, the workshops were evaluated with satisfaction, as they enabled the theoretical knowledge to be put into practice. Participants stated that they could use what they learned in the training in their professional lives and gave positive feedback on this issue with 4.33 ± 0.80 points. In general, the satisfaction level of the training was found satisfactory by the participants with 4.25 ± 0.83 points (Table 3).

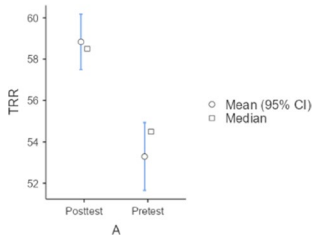
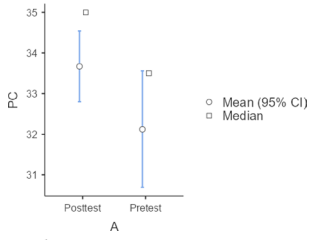
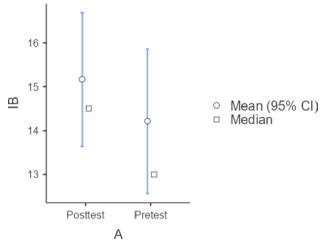
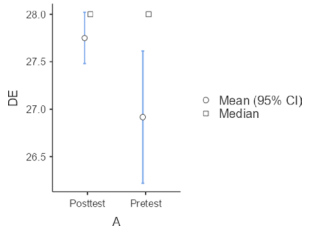
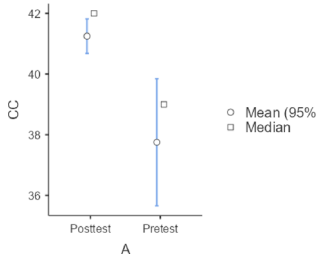
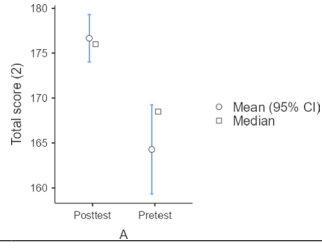
Discussion

There are many training program proposals in the literature for the development of IPC, which is a clinical approach that directly aims to improve quality in health care [7, 10, 11, 59]. This study highlighted significant improvements across all subscales of the EPIS-TR following the intervention, indicating a positive impact on interprofessional attitudes among participants. Notably, the Shapiro-Wilk test showed that the data were not normally distributed; hence, non-parametric methods were applied to analyze these results.

Findings regarding the "interprofessional belonging," "interprofessional commitment," and "interprofessional beliefs" interprofessional beliefs subscales highlight the significant impact of structured interprofessional interventions on participants' perceptions and attitudes in healthcare settings. The substantial increase in post-test scores for the "interprofessional belonging" subscale, with an effect size of 2.40, indicated that the intervention effectively fostered a sense of inclusion and community among diverse professional groups. This aligns with the notion that IPE can enhance social identity and belongingness, which are critical for effective teamwork in healthcare environments [22, 60]. Moreover, the large effect size suggests that such interventions can lead to profound changes in how professionals perceive their roles within a collaborative framework, reinforcing the importance of a shared identity in interprofessional practice [61, 62]. Similarly, the "interprofessional commitment" subscale demonstrated a statistically significant increase with a medium effect size of 0.91. This suggests that participants felt more motivated to engage in IPC, likely because of the structured interactions that clarified each profession's roles and responsibilities [63, 64]. The enhancement of commitment is crucial because it not only improves collaboration, but also contributes to better patient outcomes through coordinated care efforts [65, 66]. These findings resonate with existing literature that emphasizes the role of IPE in developing a commitment to collaborative practice among healthcare professionals [67, 68]. The improvement in the "interprofessional beliefs" subscale, with an effect size of 0.75, further supports the effectiveness of the intervention in aligning beliefs across different professions. This alignment is essential for mitigating biases and misconceptions that can impede teamwork [69, 70]. The literature suggests that fostering shared beliefs among team members can enhance mutual respect and understanding, which are vital for successful IPC [71, 72]. The overall increase in total EPIS-TR scores, with an effect size of 1.00, underscores the robust enhancement in interprofessional orientation, reinforcing the value of targeted interventions to strengthen interprofessional relationships [73, 74].

Collectively, these findings provide compelling evidence of the benefits of structured interprofessional interventions in healthcare settings. They not only enhance individual perceptions of belonging,

Table 2 IPAS analyses

	Normality Test (Shapiro-Wilk)		IPAS				
	W	***p***	**Pre test**	**Post test**		***p***	**Effect Size**
TRR (Teamwork, roles, and responsibilities)	0.94	0.015	53.25 ± 4.11	58.96 ± 3.42		< 0.001	0.70
PC (Patient-centeredness)	0.80	< 0.001	32.17 ± 3.61	33.67 ± 2.18		0.048	0.31
IB (Interprofessional bias)	0.93	0.007	14.21 ± 4.12	15.17 ± 3.81		0.403	0.14
DE (Diversity and ethics)	0.71	< 0.001	26.92 ± 1.74	27.75 ± 0.68		0.066	0.23
CC (Community centeredness)	0.78	< 0.001	37.75 ± 5.24	41.25 ± 1.42		0.004	0.44
Total score	0.93	0.005	164.29 ± 12.47	176.79 ± 6.70		< 0.001	0.65

Mann-Whitney U

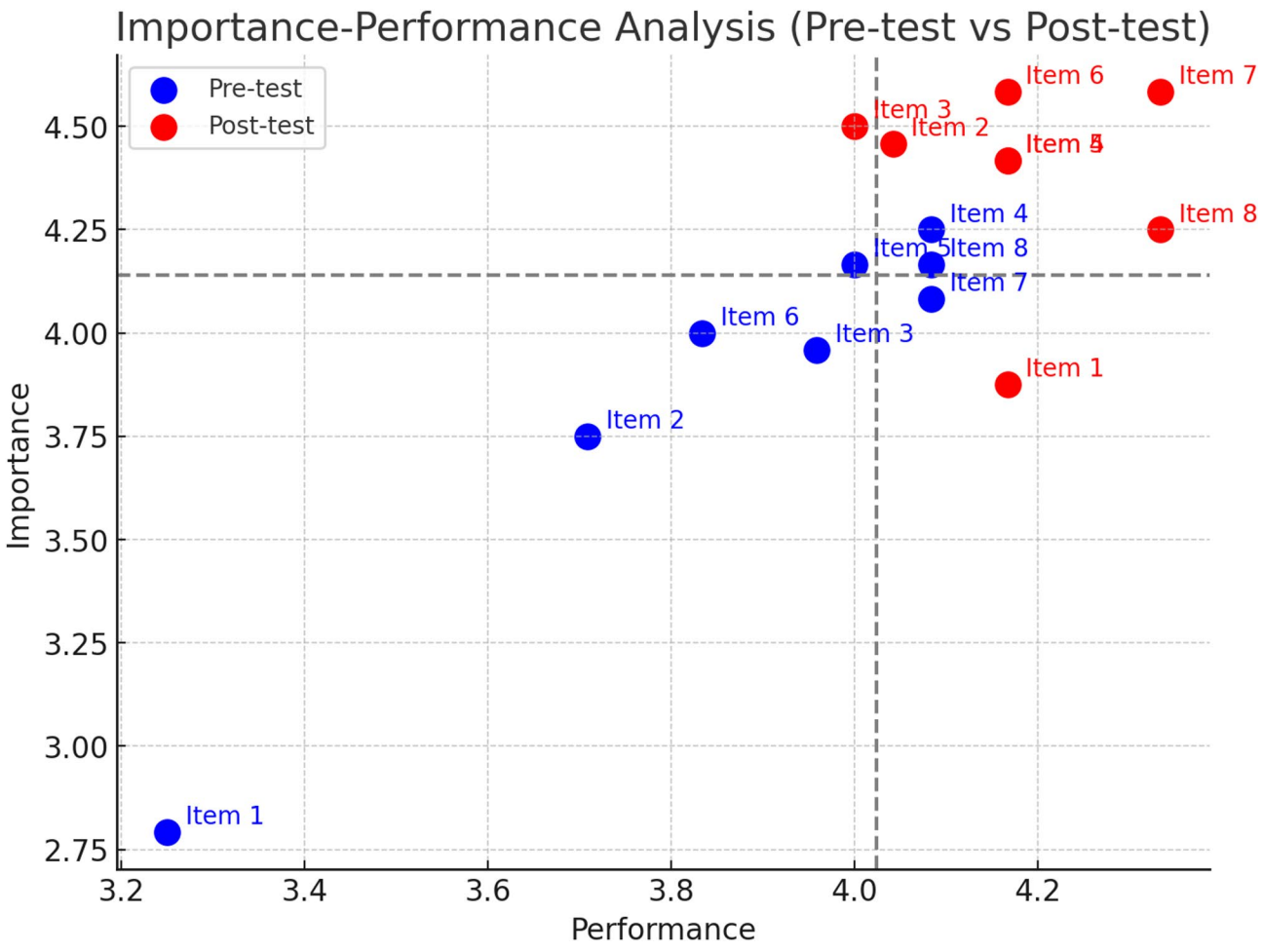


Fig. 1 Importance performance analysis

Table 3 Satisfaction analysis

	Mean	±	Standard Deviation
Organization of the training	4.33	±	0.69
Training materials	4.13	±	1.01
Lecture sessions	4.38	±	0.81
Small group training	4.04	±	0.93
Workshops	4.17	±	0.80
Ability to apply the training in my profession	4.33	±	0.80
Overall satisfaction with the training	4.25	±	0.83

commitment, and belief alignment but also lay the groundwork for future research to explore these dynamics in more diverse samples. Such research could further validate the psychometric properties of the scales used and their applicability across various healthcare contexts, ultimately contributing to the ongoing development of effective IPE strategies [75].

The results of this study provide compelling evidence of the effectiveness of the intervention in enhancing various dimensions of IPC, as measured by the IPAS. Specifically, significant increases were observed in the subscales of TRR, PC, and CC, along with an overall improvement in IPAS scores.

The TRR subscale exhibited a notable increase in post-test scores with an effect size of 0.70. This suggests a substantial enhancement in the participants' understanding and appreciation of their roles and responsibilities within interprofessional teams. These findings align with existing literature that emphasizes the importance of clearly defined roles in promoting effective teamwork and collaboration in healthcare settings [22, 61]. The intervention appears to have successfully clarified the roles of various professionals, which are essential for fostering collaboration and improving team dynamics [62, 63]. The significance of role clarity in IPE is well documented, as it can lead to improved communication, reduced conflict, and enhanced patient care outcomes [64, 65].

In the PC subscale, the intervention yielded a smaller yet statistically significant effect size of 0.31, indicating modest gains in patient-centered attitudes among participants. While the improvement in this area was less pronounced compared to the TRR subscale, it nonetheless

suggested that the intervention had a positive impact on promoting patient-centered care principles. This finding is consistent with previous studies that highlight the importance of integrating patient-centeredness into IPE [66, 67]. The modest effect size indicates that while the intervention was beneficial, there may be a need for more targeted components specifically aimed at enhancing patient-centered attitudes and practices among healthcare professionals [68, 69]. Future interventions could incorporate additional training focused on patient engagement strategies and shared decision making to further strengthen this aspect of IPC.

Although the results for the CC subscale were not explicitly detailed in the text, it is important to note that cultural competency is a critical component of effective interprofessional practice. Enhancing cultural competency among healthcare professionals is essential for addressing the diverse needs of patients and ensuring equitable care [70, 71]. Interventions that focus on cultural awareness and sensitivity can improve communication and trust between healthcare providers and patients from diverse backgrounds [72, 73]. Future research should explore the specific impact of cultural competency training in the context of IPE and its effects on patient outcomes.

The findings from this study underscore the potential of structured interprofessional interventions to enhance teamwork, roles, and responsibilities as well as to promote patient-centeredness among healthcare professionals. The significant increases in TRR and overall IPAS scores highlight the effectiveness of such interventions in fostering collaboration in healthcare settings. However, the modest gains in patient-centeredness suggest that further refinement and targeted strategies may be necessary to fully realize the potential of IPE in this area. Future research should continue to explore the nuances of these dimensions and their implications in improving IPC and patient care.

The findings indicate that while the intervention had a significant impact on certain aspects of interprofessional attitudes, it was less effective in influencing participants' interprofessional beliefs and diversity/equity awareness. The subscales for interprofessional beliefs (IB) and diversity/equity (DE) showed non-significant differences between pre- and post-tests, with small effect sizes of 0.14 and 0.23, respectively. These results suggest that additional focused strategies may be necessary to address these dimensions effectively in future interventions.

The lack of significant changes in the IB and DE subscales highlights a potential gap in the intervention's ability to shift underlying beliefs and awareness related to diversity and equity in healthcare settings. This aligns with previous research indicating that interventions aimed at enhancing cultural competency and diversity awareness often require more targeted approaches to achieve meaningful change [76, 77]. McCalman et al. emphasized the importance of organizational systems and policies in promoting cultural competence, suggesting that a comprehensive approach that includes training and community engagement is essential for improving healthcare practices [76].

In contrast, CC subscale demonstrated a statistically significant increase, with an effect size of 0.44, indicating that the intervention was effective in promoting cultural competency among participants. This finding is particularly relevant in diverse healthcare settings, where understanding cultural nuances can significantly enhance patient interactions and outcomes [78]. Research has consistently shown that cultural competence training improves health care providers' ability to engage with patients from diverse backgrounds, leading to better patient satisfaction and adherence to treatment plans [79].

The significant increase in the total IPAS scores, with an effect size of 0.65, suggests a meaningful overall improvement in interprofessional attitudes as a result of the intervention. This indicates that the program successfully strengthened participants' readiness and attitudes towards IPC, particularly in areas such as teamwork and cultural competence. The positive correlation between IPC and team effectiveness further supports the notion that enhancing inter-professional attitudes can improve outcomes in healthcare settings. While the intervention effectively promoted interprofessional attitudes and competencies, particularly teamwork and cultural competencies, the findings suggest that additional tailored strategies may be required to address interprofessional beliefs and equity-related attitudes.

The analysis revealed a statistically significant, moderately positive correlation ($r = 0.44$, $p = 0.002$) between the EPIS-TR and IPAS total scores, indicating that as participants' interprofessional belongingness (as measured by EPIS-TR) increased, their interprofessional attitudes and competencies (as measured by IPAS) also tended to improve. This relationship suggests a meaningful alignment between the constructs measured by the two scales, emphasizing that a sense of belonging within interprofessional settings may be associated with positive attitudes toward IPC.

The correlation coefficient of 0.44, while moderate, implies that EPIS-TR and IPAS share common factors that influence interprofessional attitudes and perceptions of teamwork. These findings highlight the importance of fostering strong IPI, as it may contribute to enhanced collaboration, patient-centered care, and overall professional competencies.

This study supports the interconnectedness of interprofessional belonging and attitudes, suggesting that interventions aimed at strengthening one may positively

impact the other. Future studies should further explore this relationship, perhaps by examining other contributing factors that can enhance both IPI and attitudes, ultimately fostering more collaborative healthcare environments.

Our study examined changes in participants' perceptions of IPE regarding its importance within their profession and its representation within their educational programs, both at the beginning and end of training. This training covered a comprehensive array of topics, including the significance of IPE at the national and institutional levels, IPE identity theory, roles and responsibilities in IPE, teamwork among health professionals, core IPE values, ethics, communication, and international IPE organizations.

The findings indicated an increase in participants' recognition of the importance of IPE within their profession, suggesting that training positively impacted their understanding of IPE's role in enhancing collaborative competencies and professional readiness. This heightened appreciation may reflect growing awareness of the benefits of interprofessional teamwork and the value of shared knowledge and communication skills in healthcare settings.

Conversely, there was a notable decrease in the participants' perceptions of IPE representation within their current training programs. This decline may indicate a gap between the recognized importance of IPE and the extent to which it is actively integrated into the existing curricula. The difference between perceived importance and actual implementation highlights the need for educational institutions to reassess how IPE is structured and delivered in their programs.

This program includes lectures developed within the scope of internationally accepted competencies in the field of IPC for the transformation of interprofessional identity and attitude, scenario-based small-group activities, and workshops involving participants from different professions. This application provides insight into the main structure of the program. Although this program successfully increased the perceived importance of IPE, it also highlighted a significant area of development in its integration into formal education. This gap may require future program development initiatives that include interprofessional learning experiences of health professionals. Future comprehensive studies should strengthen the role of IPE in the program and evaluate the long-term impact of these activities on professional practice and patient outcomes.

The impact of IPC on the quality of healthcare services has frequently been discussed. The training program developed in this national study provides evidence for future inter-professional collaboration studies. This study, which is insightful in terms of future perspectives on interprofessional collaboration, is also the first to claim that students' perceptions of the importance of IPC increase with training and that they become aware of its place/lack in their own training programs. The limitations of this study include the relatively small number of participants, voluntary participation, female participants, and lack of long-term results. However, it is valuable in terms of showing the impact of an educational program on the identity and attitudes of IPC. The program developed in this study is an important reference for educational programs aimed at developing IPC, in terms of its structure and effectiveness. We believe that this application can contribute to the integration of IPC into health professional education programs.

Abbreviations

IPC	Interprofessional collaboration
IPI	Interprofessional identity
IPA	Interprofessional attitude
IPE	Interprofessional education
EPIS-TR	Extended Professional Identity Scale
IPAS	Interprofessional Attitude Scale
TRR	Teamwork, roles, and responsibilities
PC	Patient-centeredness
IPB	Interprofessional bias
DE	Diversity and ethics
CC	Community centeredness

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

GK: Conceptualization; Data collection, formal analysis, investigation, methodology, validation, review, and editing.MİBK: Conceptualization; Project administration, data curation, formal analysis, investigation, methodology, validation, visualization, writing the original draft.MP: Conceptualization; Formal analysis, investigation, methodology, resources, supervision, validation, reviewing, and editing.All authors reviewed the manuscript.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Ethics committee approval was obtained from the "Süleyman Demirel University Ethics Committee" (Date: 03.07.2024 No: 77/2). In this study, consent that was obtained from all of the participants was informed.

Consent for publication

Not Applicable.

Competing interests
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